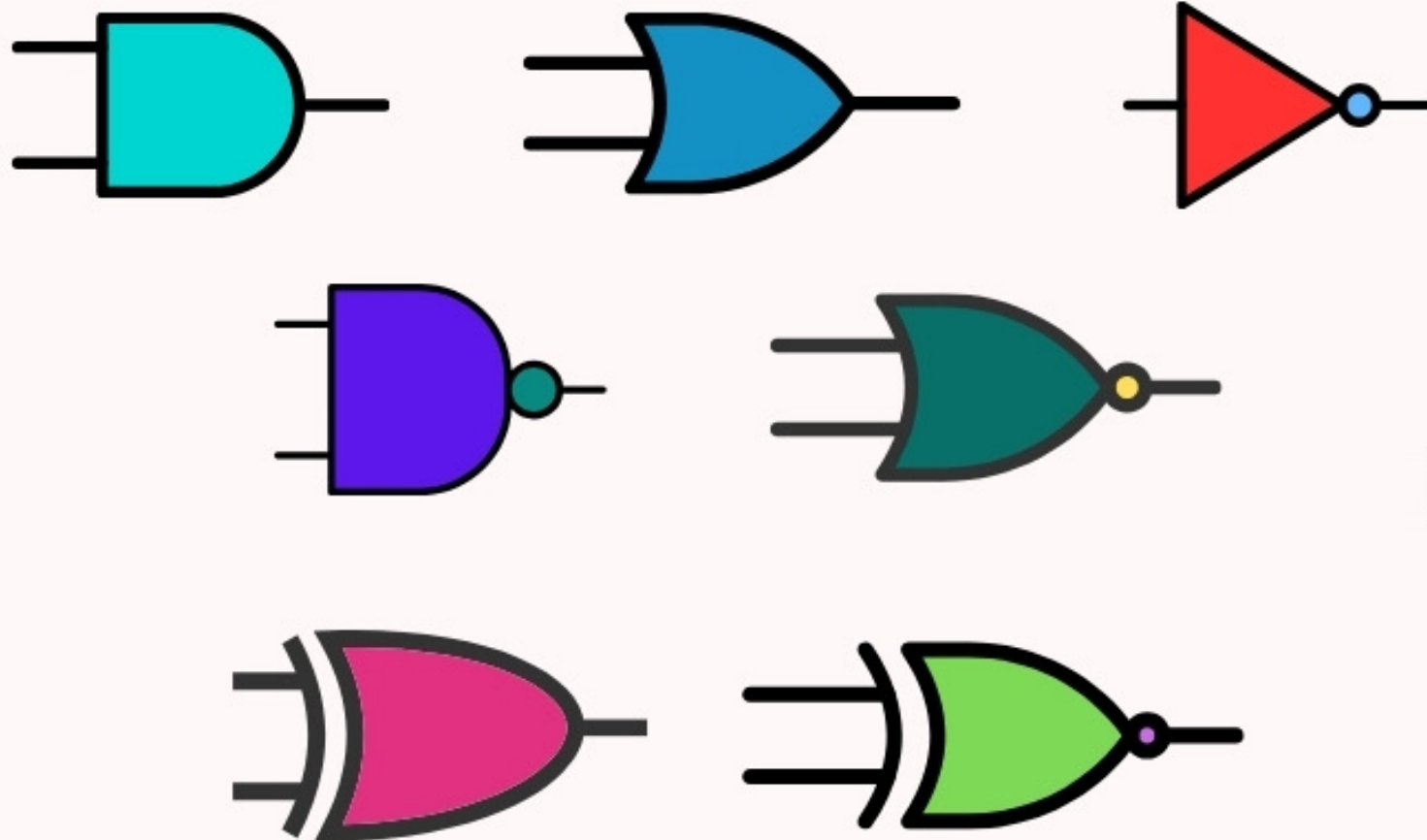


Gene Expression Logic Gates

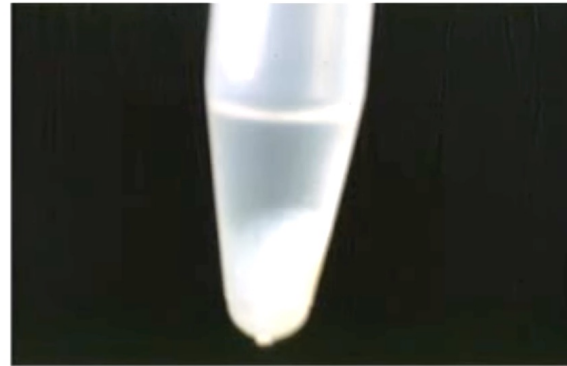


Dissection of DNA

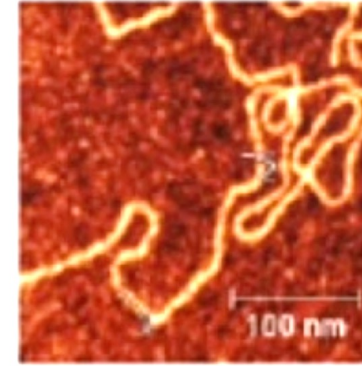
10,000,000,000
(10^6) 1 Billion
Bacteria



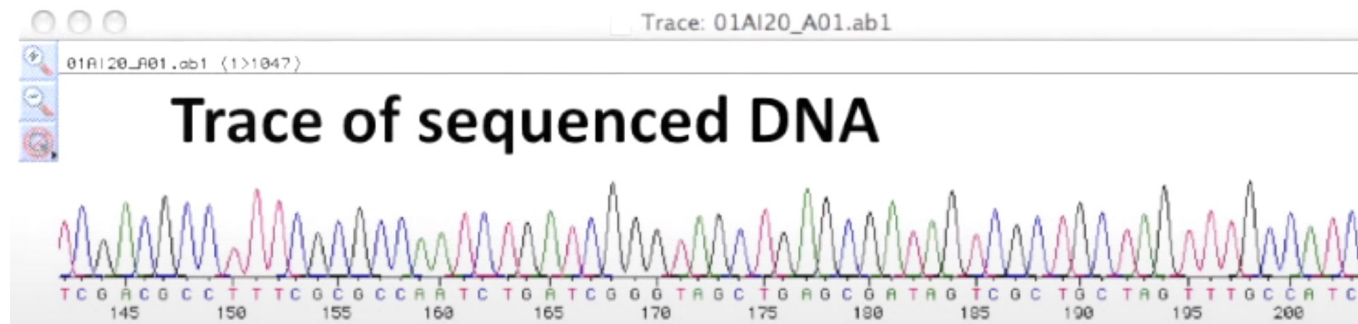
**culture
of cells**



isolation of DNA



**DNA seen
by AFM**



Trace of sequenced DNA

Codes of cell
functions

Lac Operon

Gene z: Galactosidase

Lactose- Glucose

Gene y: Permease

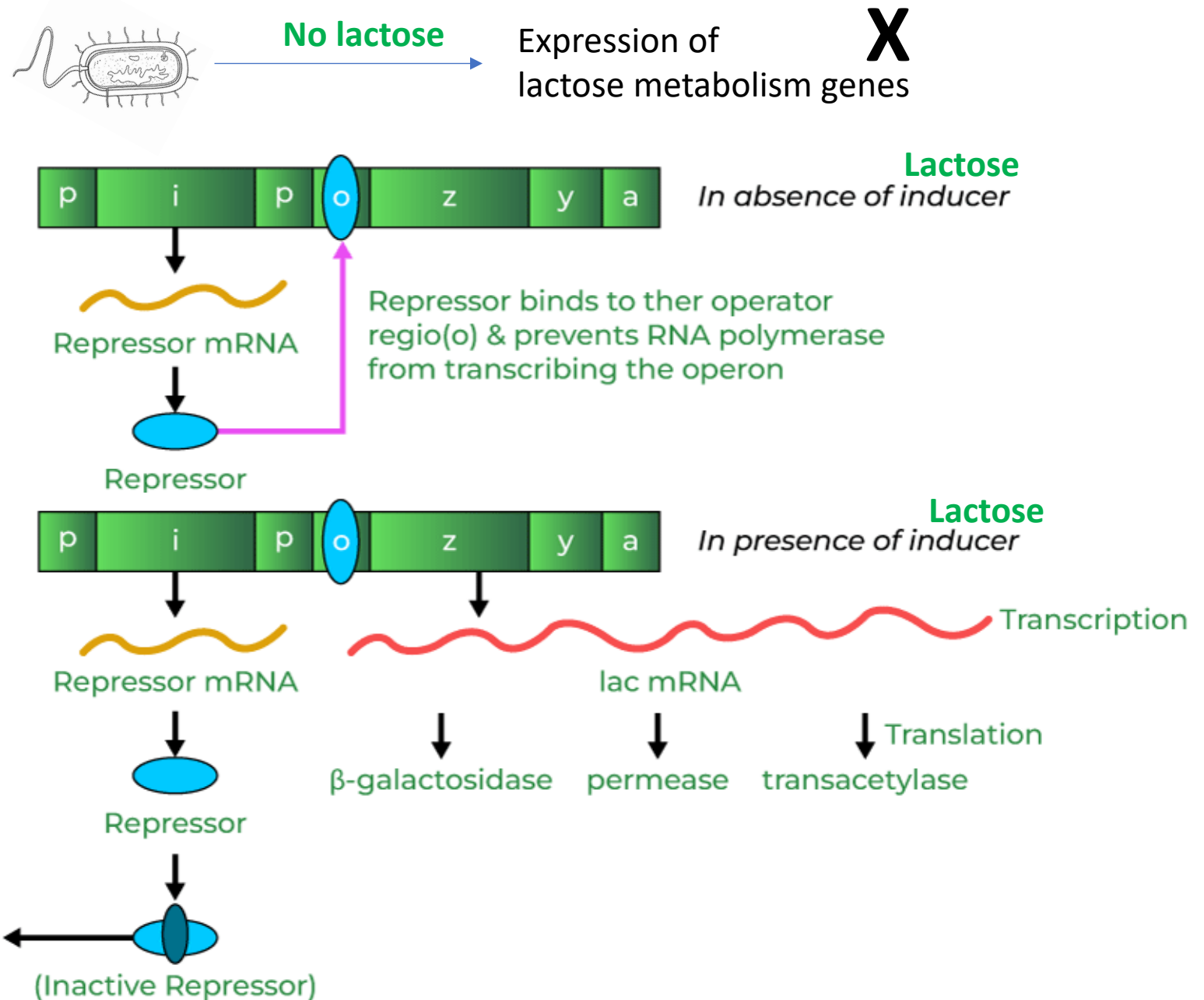
Lactose crossing membrane

Gene a: Transacetylase

Helps Galactosidase



Lactose



Sequence of E.Coli Genome

- 6×10^6 nucleotides
- 2000 characters per page
- 3000 pages for the entire genome
- The Human Genome is 1000 times bigger

What does that sequence mean?

```
1 Thr Met Ile Thr Asp Ser Leu Ala Val Val Leu Gln Arg Arg Asp Trp Glu Asn Pro Gly
1 ACC ATG ATT ACG GAT TCA CTG GCC GTC GTT TTA CAA CGT CGT GAC TGG GAA AAC CCT GGC

21 Val Thr Gln Leu Asn Arg Leu Ala Ala His Pro Pro Phe Ala Ser Trp Arg Asn Ser Glu
61 GTT ACC CAA CTT AAT CGC CTT GCA GCA CAT CCC CCT TTC GCC AGC TGG CGT AAT AGC GAA

41 Glu Ala Arg Thr Asp Arg Pro Ser Gln Gln Leu Arg Ser Leu Asn Gly Glu Trp Arg Phe
121 GAG GCC CGC ACC GAT CGC CCT TCC CAA CAG TTG CGC GAA TGG CGC GAT TTT

61 Ala Trp Phe Pro Ala Pro Glu Ala Val Pro Glu Ser Trp Leu Glu Cys Asp Leu Pro Glu
181 GCC TGG TTT CCG GCA CCA GAA GCG GTG CCG GAA AGC TGG CTG GAG TGC GAT CTT CCT GAG

81 Ala Asp Thr Val Val Val Pro Ser Asn Trp Gln Met His Gly Tyr Asp Ala Pro Ile Tyr
241 GCC GAT ACT GTC GTC GTC CCC TCA AAC TGG CAG ATG CAC GGT TAC GAT GCG CCC ATC TAC

101 Thr Asn Val Thr Tyr Pro Ile Thr Val Asn Pro Pro Phe Val Pro Thr Glu Asn Pro Thr
301 ACC AAC GTA ACC TAT CCC ATT ACG GTC AAT CCG CCG TTT GTT CCC ACG GAG AAT CCG ACG

121 Gly Cys Tyr Ser Leu Thr Phe Asn Val Asp Glu Ser Trp Leu Gln Glu Gly Gln Thr Arg
361 GGT TGT TAC TCG CTC ACA TTT AAT GTT GAT GAA AGC TGG CTA CAG GAA GGC CAG ACG CGA

141 Ile Ile Phe Asp Gly Val Asn Ser Ala Phe His Leu Trp Cys Asn Gly Arg Trp Val Gly
421 ATT ATT TTT GAT GGC GTT AAC TCG GCG TTT CAT CTG TGG TGC AAC GGT TGG GTG GGT

161 Tyr Gly Gln Asp Ser Arg Leu Pro Ser Glu Phe Asp Leu Ser Ala Phe Leu Arg Ala Gly
481 TAC GGC CAG GAC AGT CGT TTG CCG TCT GAA TTT GAC CTG AGC GCA TTT TTA CGC GCC GGA

181 Glu Asn Arg Leu Ala Val Met Val Leu Arg Trp Ser Asp Gly Ser Tyr Leu Glu Asp Gln
541 GAA AAC CGC CTC GCG GTG ATG GTG CTG CGT TGG AGT GAC GGC AGT TAT CTG GAA GAT CAG

201 Asp Met Trp Arg Met Ser Gly Ile Phe Arg Asp Val Ser Leu Leu His Lys Pro Thr Thr
601 GAT ATG TGG CCG ATG AGC GGC ATT TTC CGT GAC GTC TCG TTG CTG CAT AAA CCG ACT ACA

221 Gln Ile Ser Asp Phe His Val Ala Thr Arg Phe Asn Asp Asp Phe Ser Arg Ala Val Leu
661 CAA ATC AGC GAT TTC CAT GTT GCC ACT CGC TTT AAT GAT GAT TTC AGC CGC GCT GTA CTG

241 Glu Ala Glu Val Gln Met Cys Gly Glu Leu Arg Asp Tyr Leu Arg Val Thr Val Ser Leu
721 GAG GCT GAA GTT CAG ATG TGC GGC GAG TTG CGT GAC TAC CTA CGG GTA ACA GTT TCT TTA

261 Trp Gln Gly Glu Thr Gln Val Ala Ser Gly Thr Ala Pro Phe Gly Gly Glu Ile Ile Asp
781 TGG CAG GGT GAA ACG CAG GTC GCC AGC GGC ACC GCG CCT TTC GGC GGT GAA ATT ATC GAT

281 Glu Arg Gly Gly Tyr Ala Asp Arg Val Thr Leu Arg Leu Asn Val Glu Asn Pro Lys Leu
841 GAG CGT GGT GGT TAT GCC GAT CGC GTC ACA CTA CGT CTG AAC GTC GAA AAC CCG AAA CTG

301 Trp Ser Ala Glu Ile Pro Asn Leu Tyr Arg Ala Val Val Glu Leu Thr Thr Ala Asp Gly
901 TGG AGC GCC GAA ATC CCG AAT CTC TAT CGT GCG GTG GTT GAA CTG CAC ACC GCC GAC GGC

321 Thr Leu Ile Glu Ala Glu Ala Cys Asp Val Gly Phe Arg Glu Val Arg Ile Glu Asn Gly
961 ACG CTG ATT GAA GCA GAA GCC TGC GAT GTC GGT TTC CGC GAG GTG CCG ATT GAA AAT GGT

341 Leu Leu Leu Leu Asn Gly Lys Pro Leu Leu Ile Arg Gly Val Asn Arg His Glu His His
1021 CTG CTG CTG CTG AAC GGC AAG CCG TTG CTG ATT CGA GGC GTT AAC CGT CAC GAG CAT CAT

361 Pro Leu His Gly Gln Val Met Asp Glu Gln Thr Met Val Gln Asp Ile Leu Leu Met Lys
1081 CCT CTG CAT GGT CAG GTC ATG GAT GAG CAG ACG ATG GTG CAG GAT ATC CTG CTG ATG AAG

381 Gln Asn Asn Phe Asn Ala Val Arg Cys Ser His Tyr Pro Asn His Pro Leu Trp Tyr Thr
1141 CAG AAC AAC TTT AAC GCC GTG GCG TGT TCG CAT TAT CCG AAC CAT CCG CTG TGG TAC ACG

401 Leu Cys Asp Arg Tyr Gly Leu Tyr Val Val Asp Glu Ala Asn Ile Glu Thr His Gly Met
1201 CTG TGC GAC CGC TAC GGC CTG TAT GTG GTG GAT GAA GCC AAT ATT GAA ACC CAC GGC ATG

421 Val Pro Met Asn Arg Leu Thr Asp Asp Pro Arg Trp Leu Pro Ala Met Ser Glu Arg Val
1261 GTG CCA ATG AAT CGT CTG ACC GAT GAT CCG GCG TGG CTA CCG GCG ATG AGC GAA CCG GTA

441 Thr Arg Met Val Gln Arg Arg Asn His Pro Ser Val Ile Ile Trp Ser Leu Gly Asn
1321 ACG CGA ATG GTG CAG CGC GAT CGT AAT CAC CCG AGT GTG ATC ATC TGG TCG CTG GGG AAT

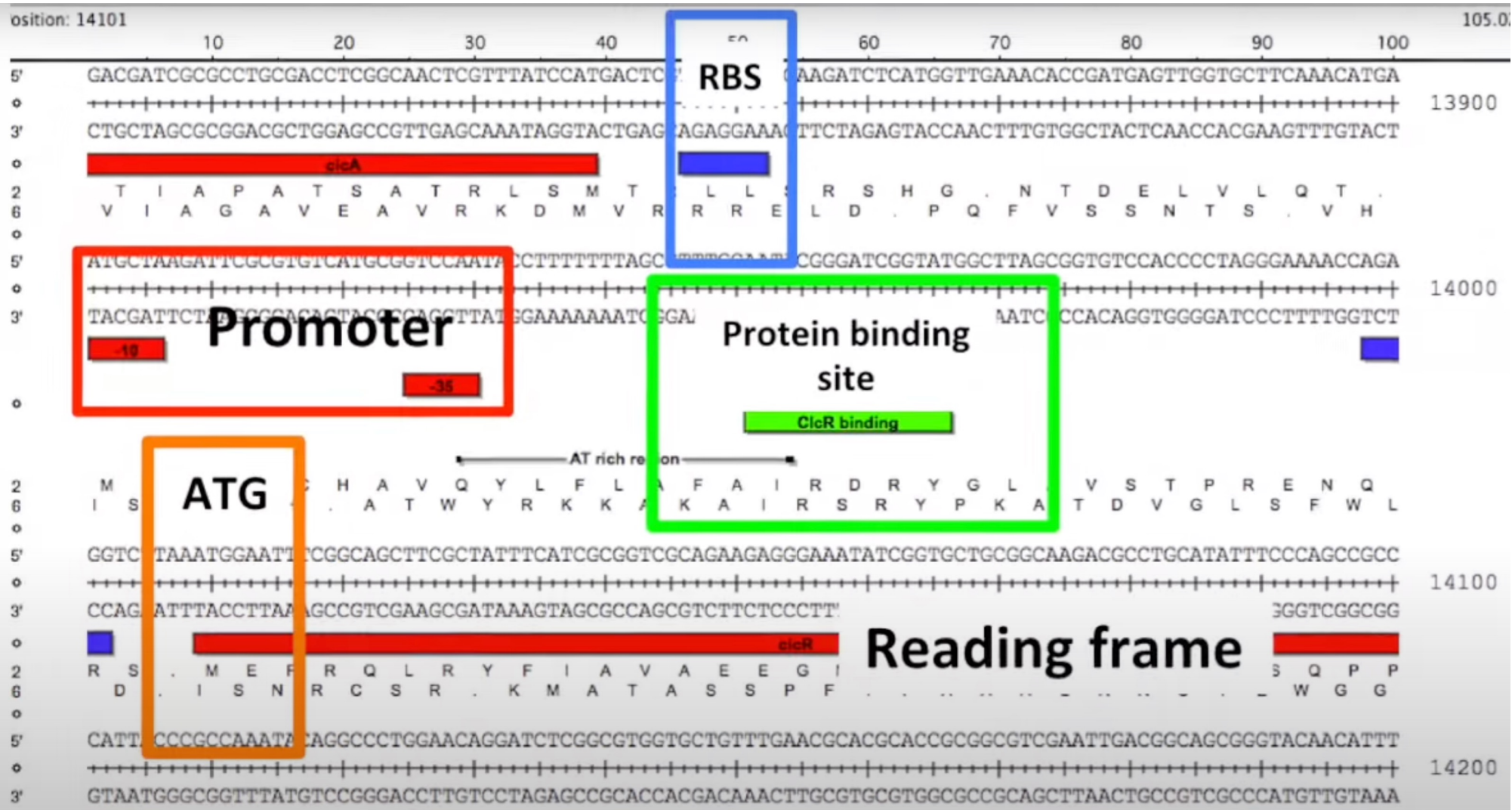
461 Glu Ser Gly His Gly Ala Asn His Asp Ala Leu Tyr Arg Trp Ile Lys Ser Val Asp Pro
1381 GAA TCA GGC CAC GGC GCT AAT CAC GAC GCG CTG TAT CGC TGG ATC AAA TCT GTC GAT CCT

481 Ser Arg Pro Val Gln Tyr Glu Gly Gly Ala Asp Thr Thr Thr Thr Thr Thr Thr Thr
1441 TCC CGC CCG GTG CAG TAT GAA GGC GGC GGA ACC GAC ACC ACG GCC ACC GAT ATT ATT TGC

501 Pro Met Tyr Ala Arg Val Asp Glu Asp Gln Pro Phe Pro Ala Val Pro Lys Trp Ser Ile
1501 CCG ATG TAC GCG CGC GTG GAT GAA GAC CAG CCC TTC CCG GCT GTG CCG AAA TGG TCC ATC

521 Lys Lys Trp Leu Ser Leu Pro Gly Glu Thr Arg Pro Leu Ile Leu Cys Glu Tyr Ala His
1561 AAA AAA TGG CTT TCG CTA CCT GGA GAG ACG GCG CCG CTG ATC CTT TGC GAA TAC GCC CAC
```

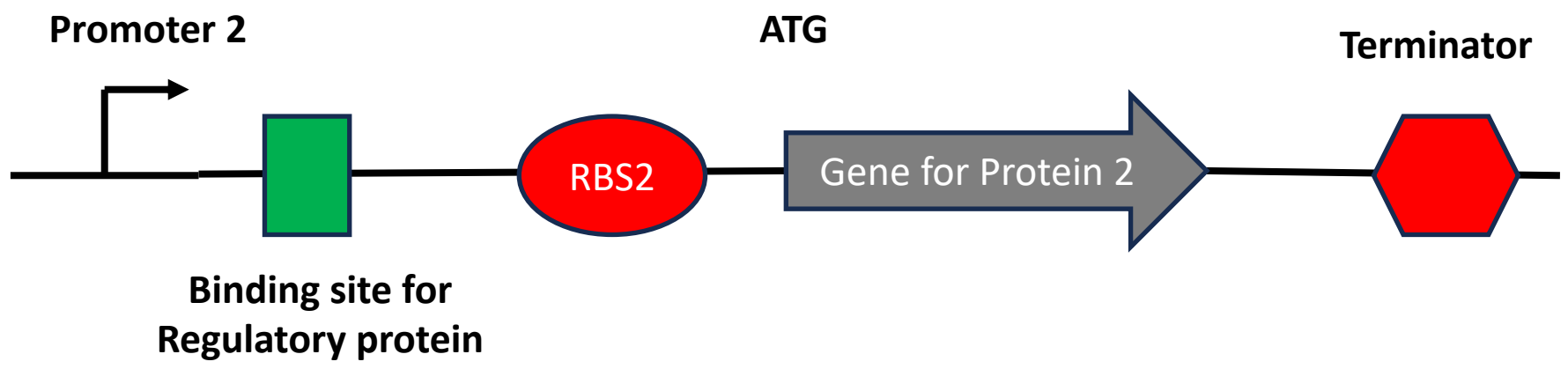
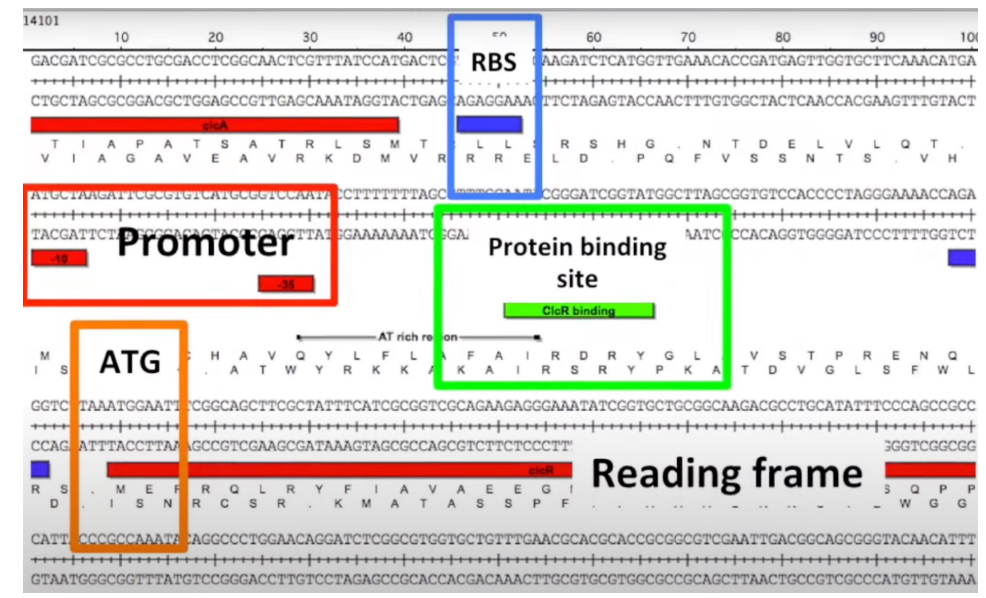
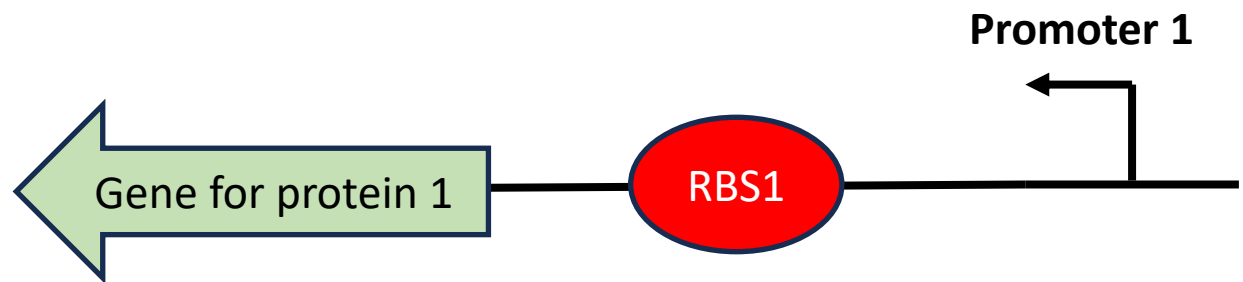

Sequence analysis



Different parts of
DNA sequence
have different
functions

Can act as
Ribosome binding
sites, promoters,
protein-coding
frames, Protein
binding sites

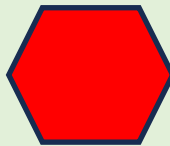
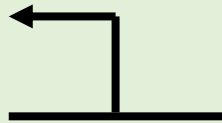
From DNA Sequence to “Circuit”



These DNA parts are arranged as a circuit

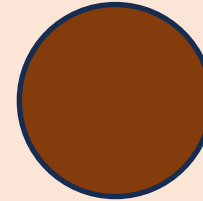
Circuit parts

DNA construction bricks

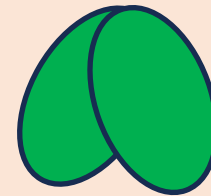


Protein parts

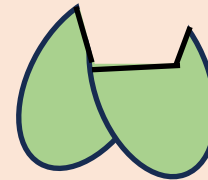
Structural
protein



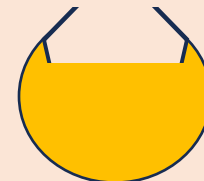
Regulator



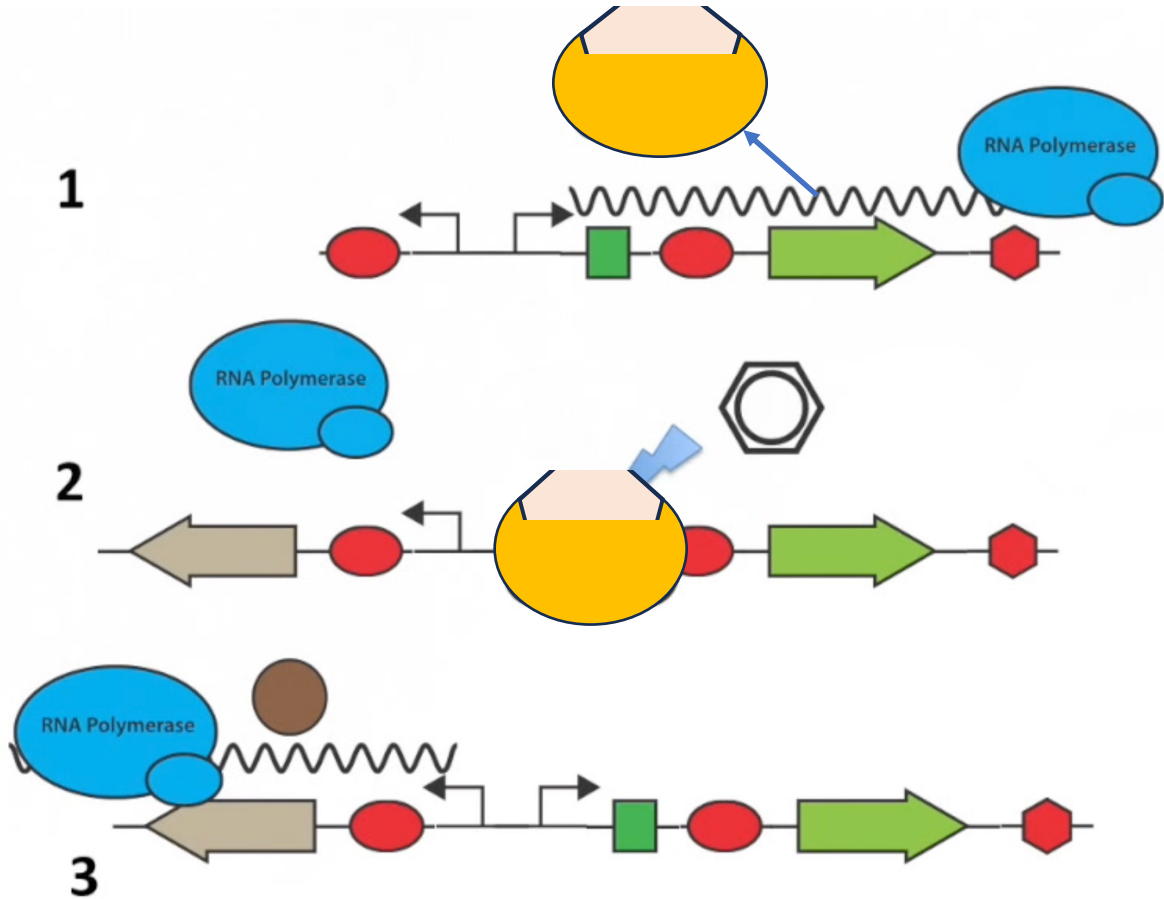
Transcription
factor



Sensory
protein



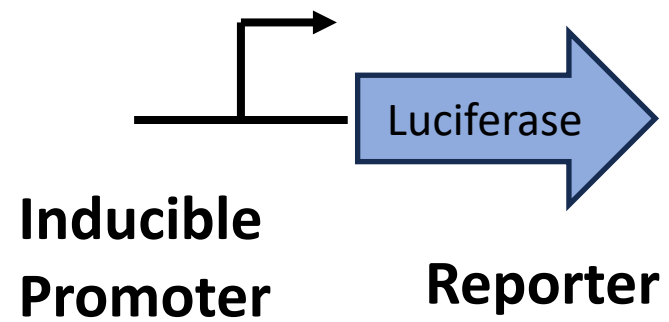
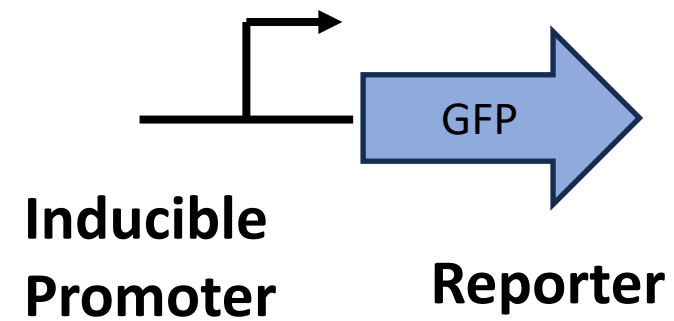
Rules and models



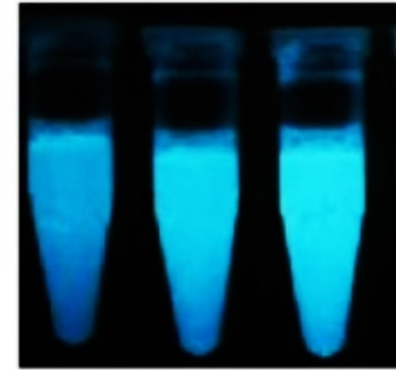
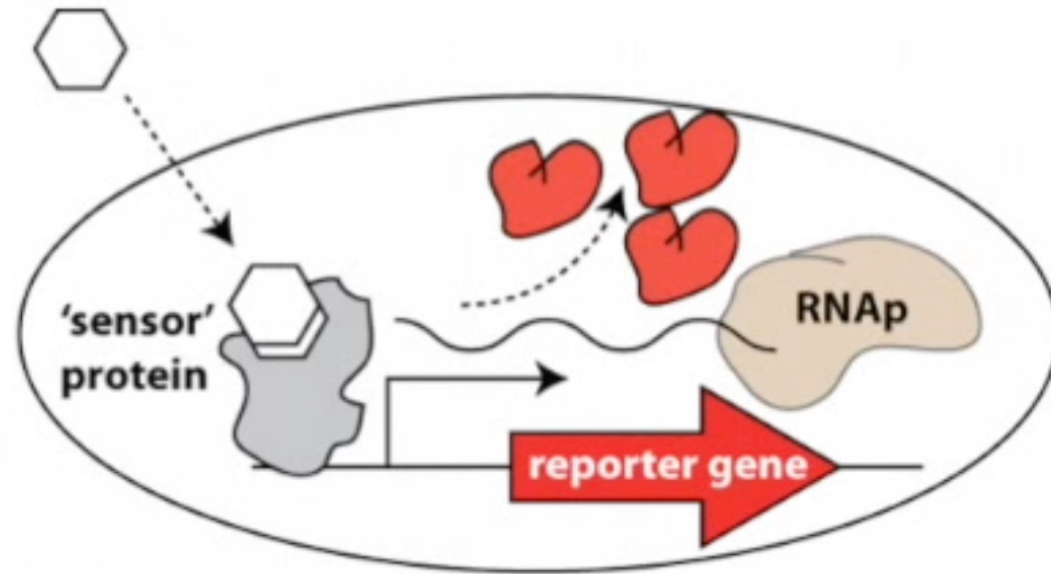
A part of the circuit synthesizes sensory protein

Sensory protein binds to a promoter of another gene

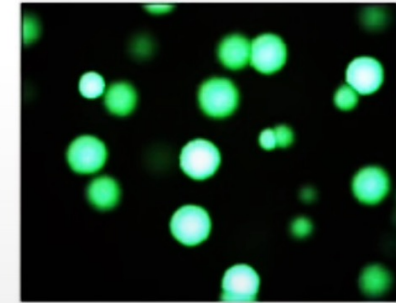
Sensory protein attracts RNA Polymerase to transcribe



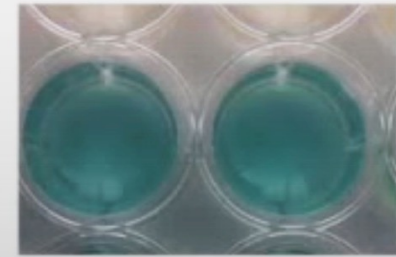
Bioreporters



Bioluminescence

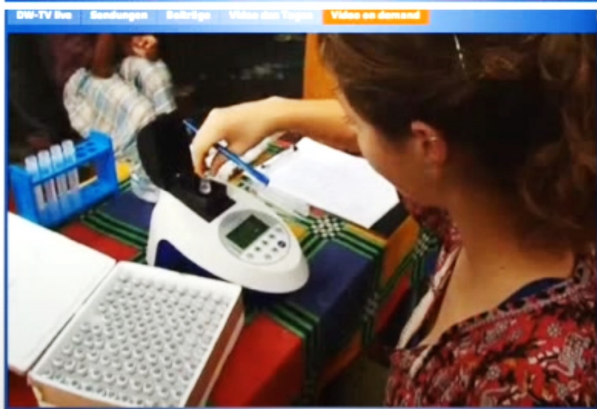


Fluorescence



Colors

Bioreporters for environment

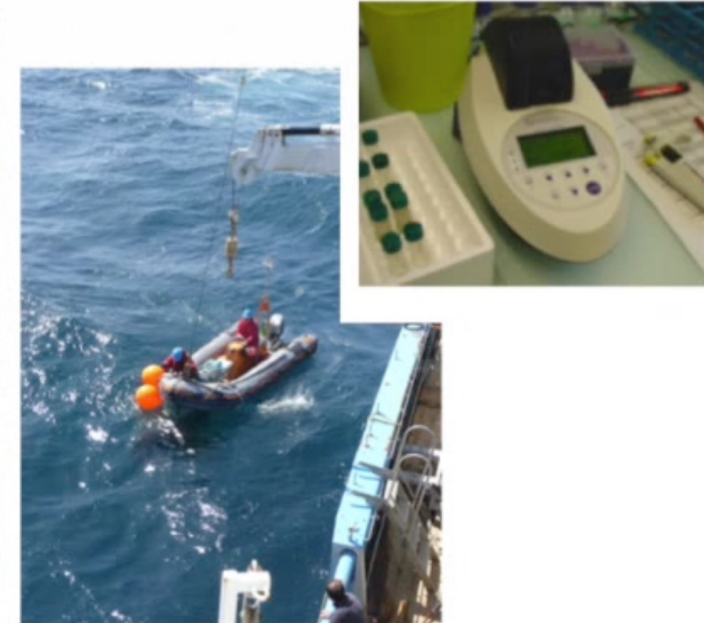


**ARSOLUX-system.
Collaboration with
UFZ-Leipzig**



courtesy: André Künzelmann

Bild: André Künzelmann

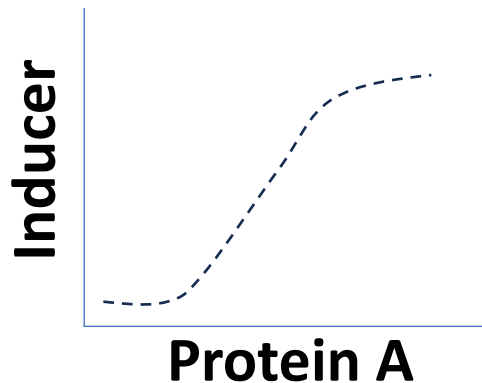
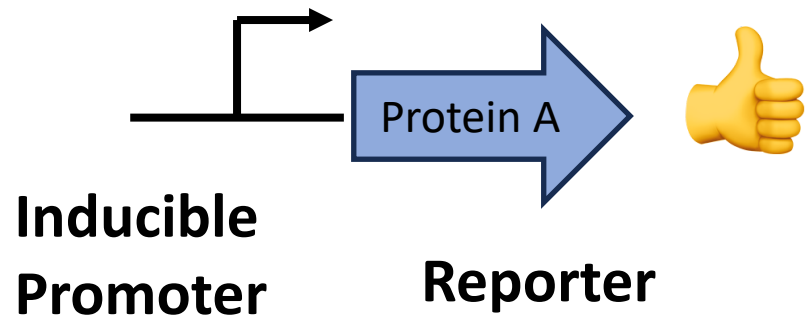


**2008, 2009, with permission of
the Dutch Government and help
of the responder vessel MSS Arca**

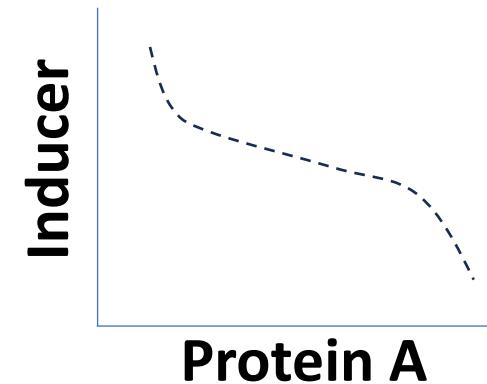
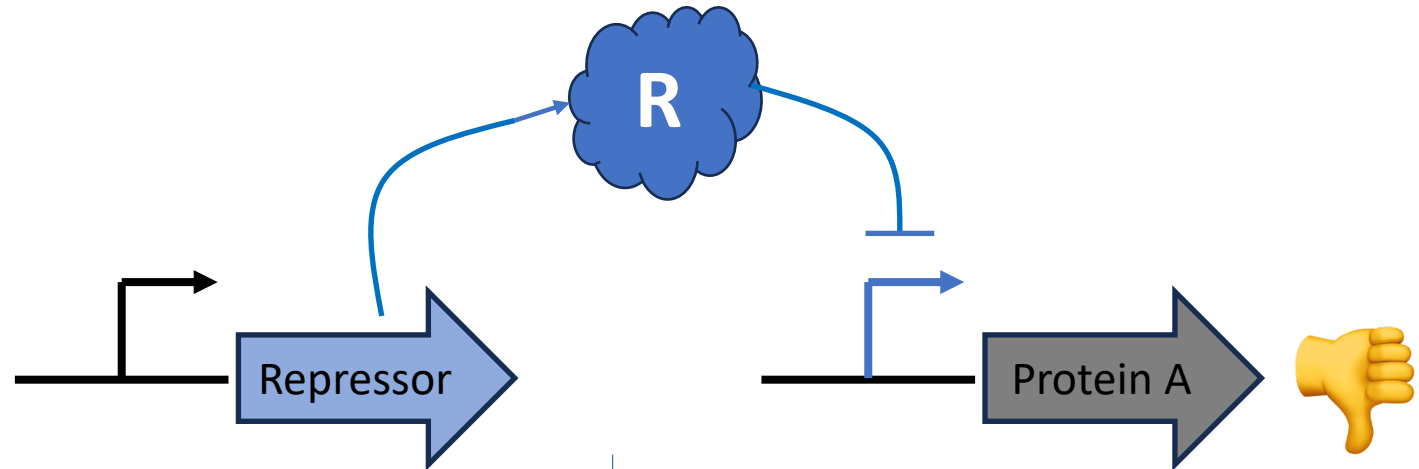
Transcriptional circuits

- It consists of nodes (DNA parts) and edges represent their connectivity

Constitutive expression

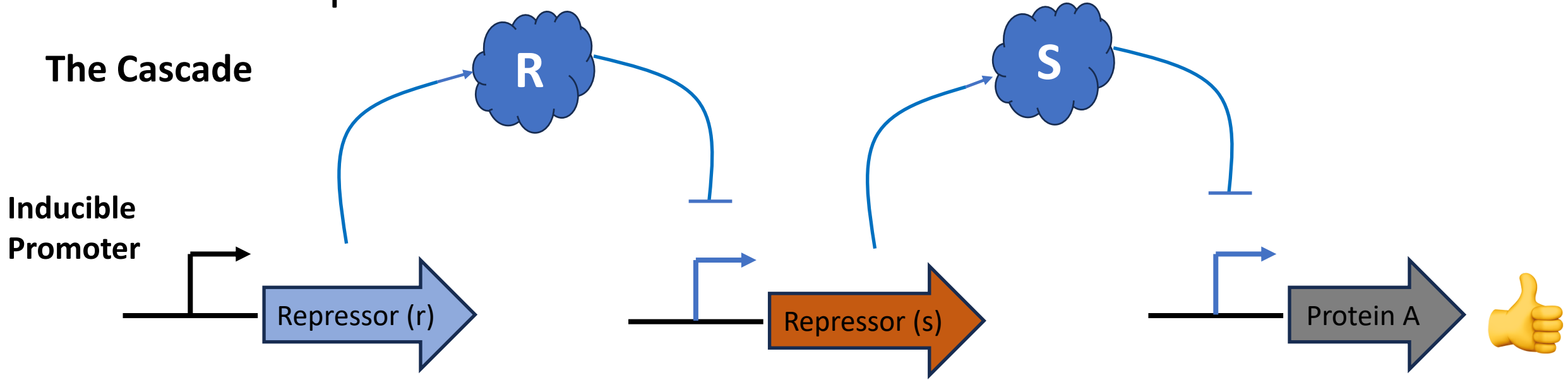


The Inverter



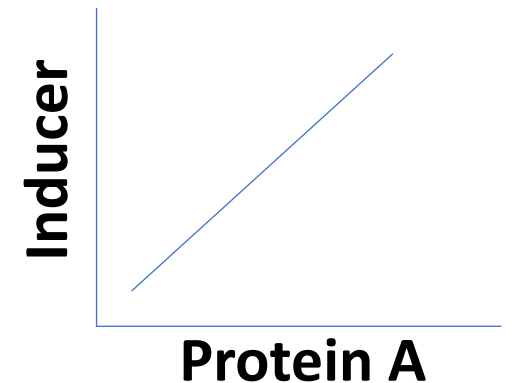
Transcriptional circuits

The Cascade



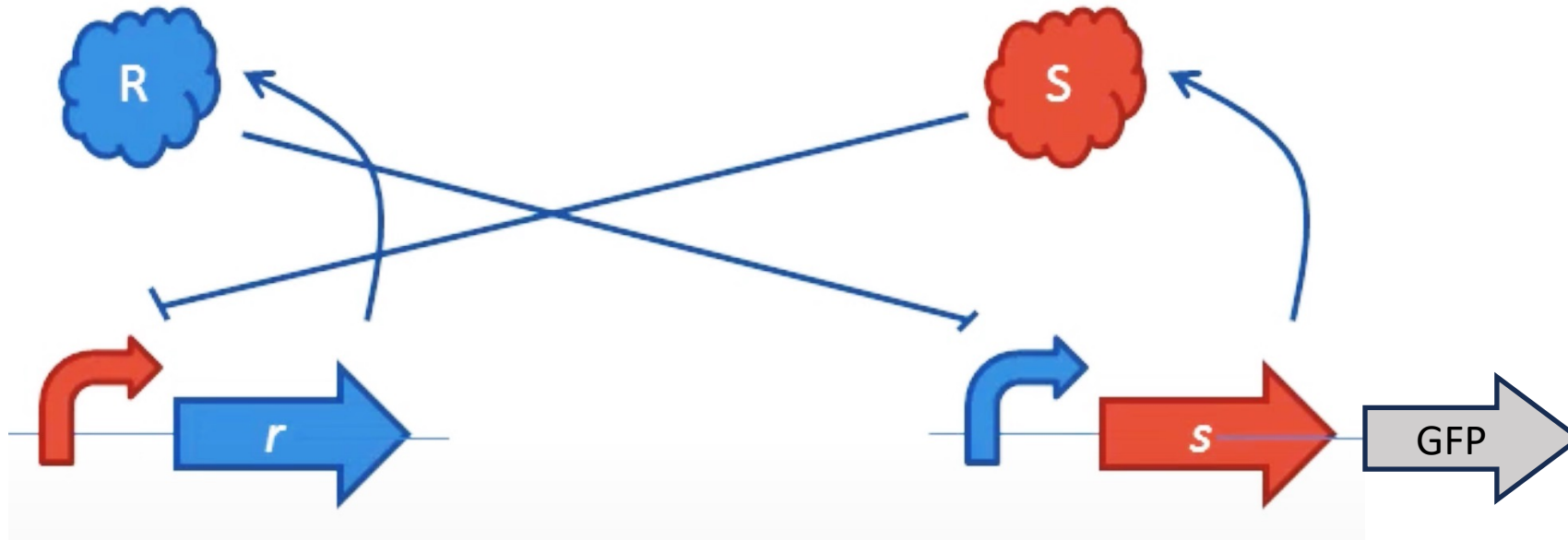
Repressor = 0	repressor = 1	repressor = 2	repressor = 3	repressor = 4
👍	👎	👍	👎	👍

Repressor	Response
10	👍
7	👎
13	👎



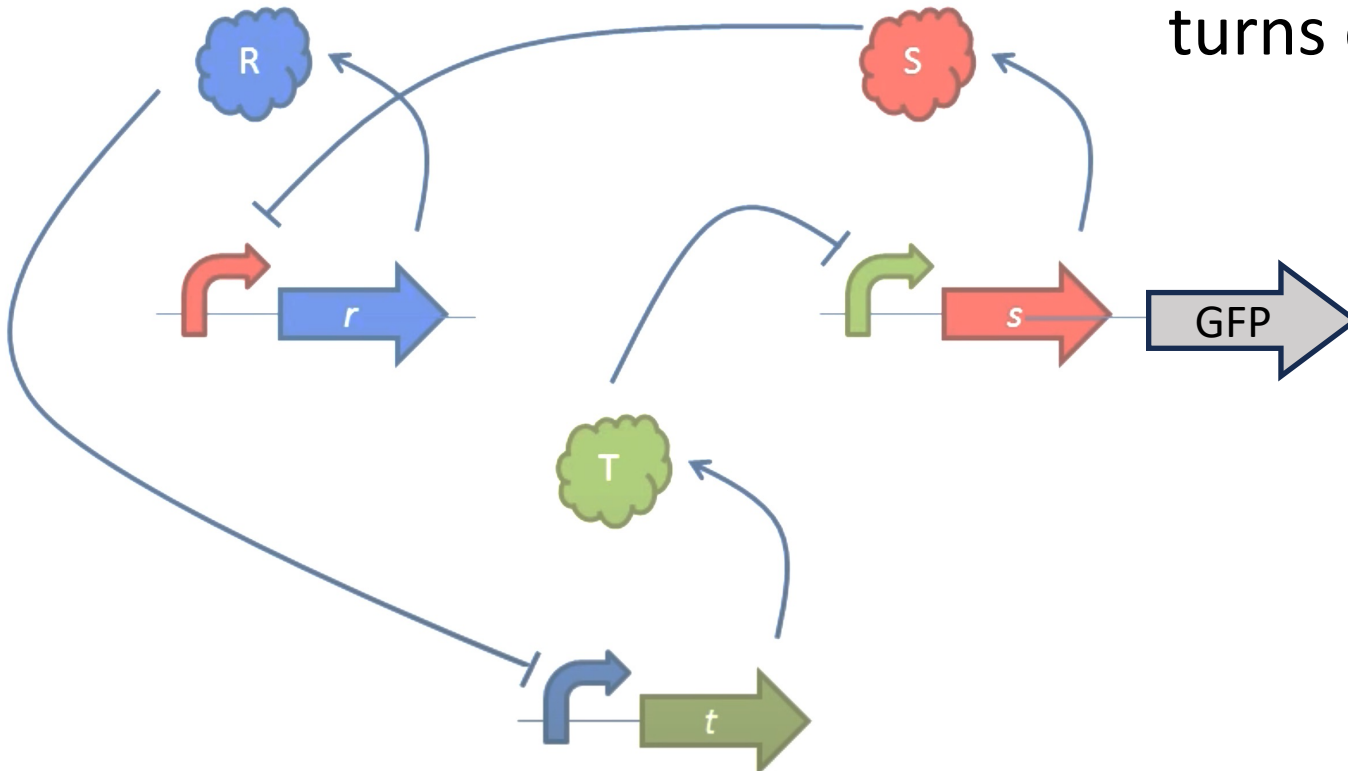
Transcriptional circuits

The toggle switch



The Repressilator

It is an oscillator where one gene turns off the other



k

knowable PRESENTS



Cellular Rhythms: A Rock-Paper-Scissors Circuit

<https://www.youtube.com/watch?v=EHPUa4sztuE&t=131s>

Engineering idea

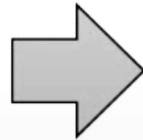
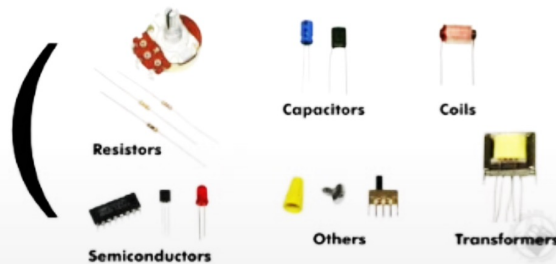
Biological parts



Construction
and verification



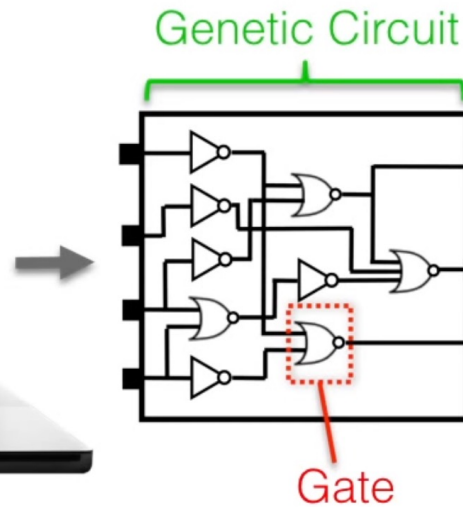
Rules, models,
engineering



[http://eroelectronic.net/
electronic-circuit-board/](http://eroelectronic.net/electronic-circuit-board/)

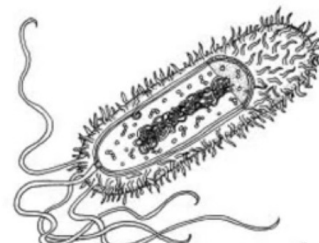
Schematic of cell programming

Write desired circuit on computer...

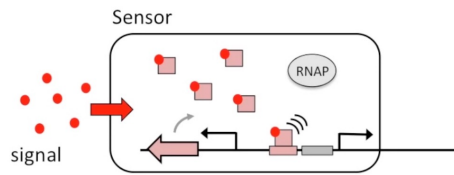


... convert to DNA sequence

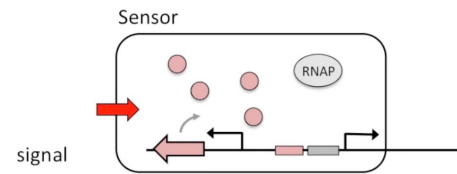
```
ACATTGATTATTTGACGGCGTCACACTTTGCTATGCCATAGC
ATTTTTATCCATAAGATTAGCGGATCCTACCTGACGCTTTTA
TCGCAACTCTCTACTGTTTCTCCATACCGTTTTTTGGGCTAG
CTACTAGAGAAAGAGGAGAAATACTAGATGAGTGTCACACTTAG
CTTCCCAGTTGCGACATTGATTATTTGACGGCGTCACACTTT
GCTATGCCATAGCATTTTTATCCATAAGATTAGCGGATCCTAC
CTGACGCTTTTTATCGCAACTCTCTACTGTTTCTCCATACCGT
TTTTTTGGGCTAGCTACTAGAGAAAGAGGAGAAATACTAGATG
AGTGTCAACTTAGCTTCCCAGTTGCGACATTGATTATTTGAC
GGCGTCACACTTTGCTATGCCATAGCATTTTTATCCATAAGAT
TAGCGGATCCTACCTGACGCTTTTTATCGCAACTCTCTACTGT
TTCTCCATACCGTTTTTTTGGGCTAGCTACTAGAGAAAGAGGA
GAAATACTAGATGAGTGTCACACTTAGCTTCCCAGTTGCGACAT
TGATTATTTGACGGCGTCACACTTTGCTATGCCATAGCATT
TTATCCATAAGATTAGCGGATCCTACCTGACGCTTTTTATCGC
AACTCTCTACTGTTTCTCCATACCGTTTTTTTGGGCTAGCTAC
TAGAGAAAGAGGAGAAATACTAGATGAGTGTCACACTTAGCTTC
CCAGTTGCG... 2-10kb of DNA
```



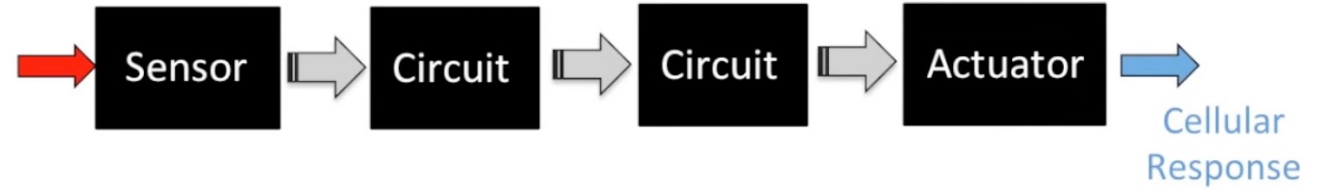
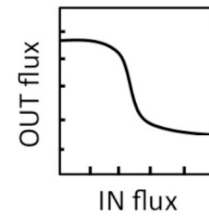
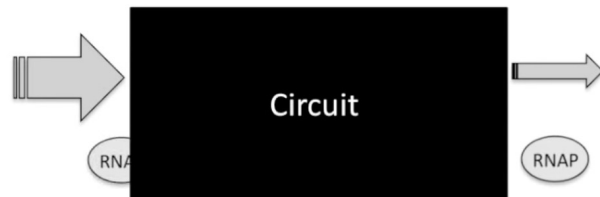
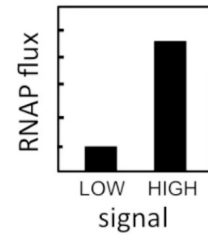
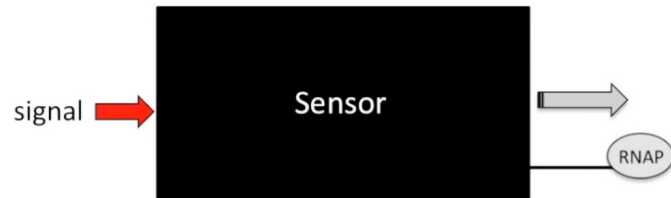
- The user writes a program
- The software converts into a circuit
- The software converts the circuit into the DNA sequence
- A company can synthesize the DNA



High



Low



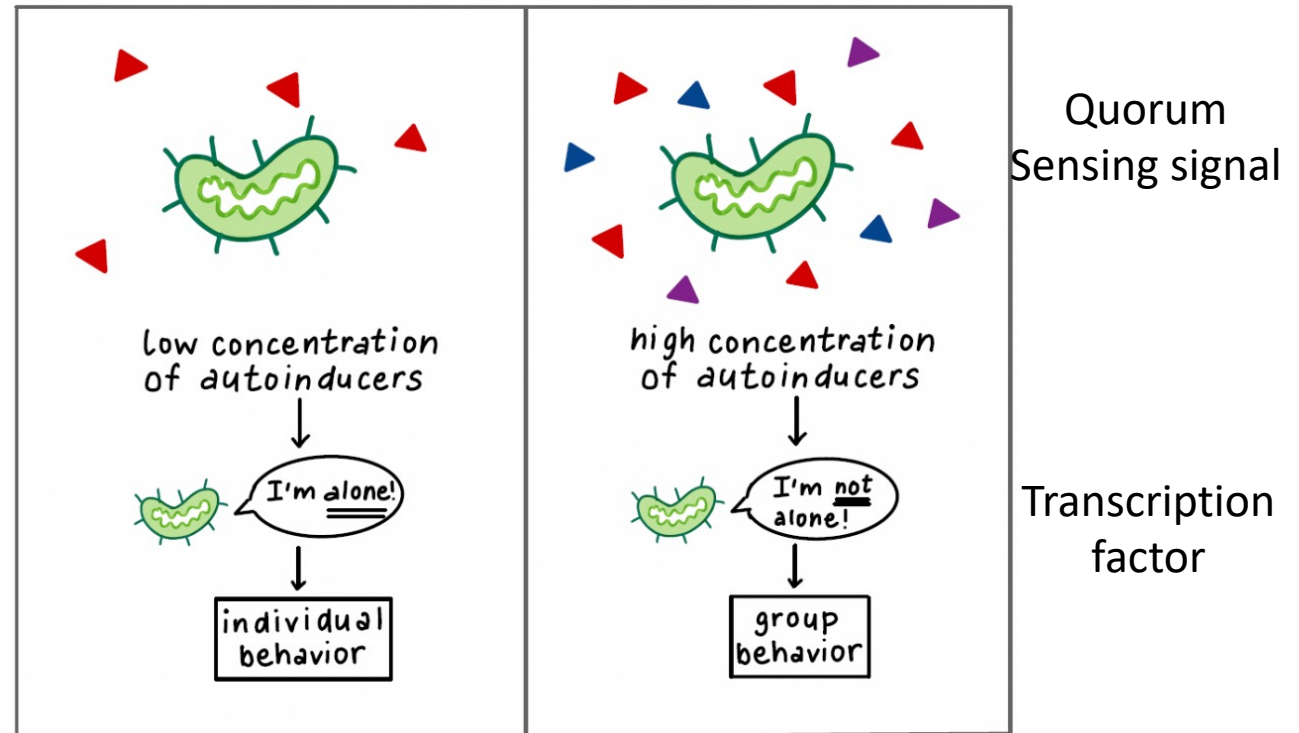
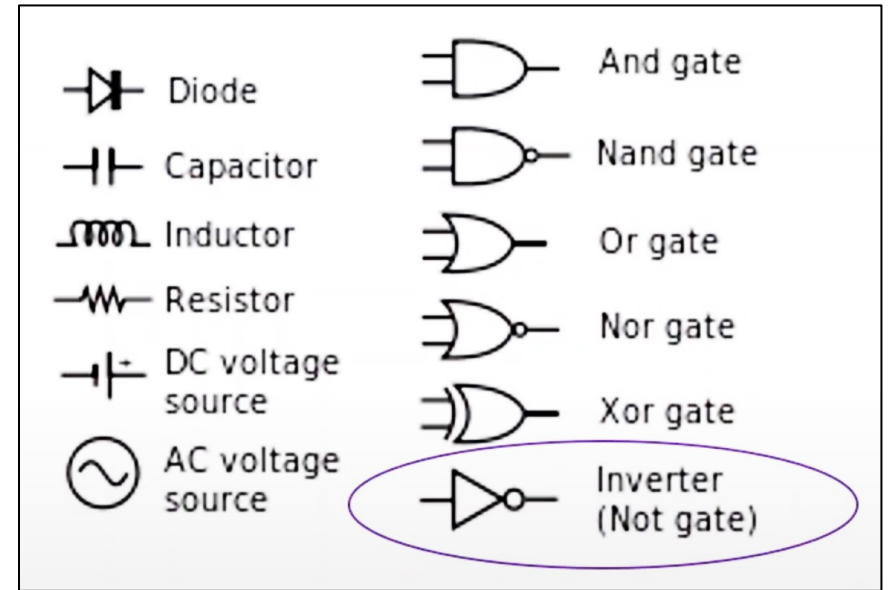
Engineering genetic circuits

Why to design new genetic circuits?

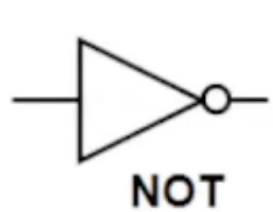
Create/program new behaviors in the cell (Synthetic Biology)

Consider this: $A \longrightarrow C$

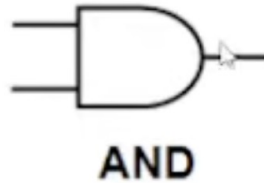
Logic gates



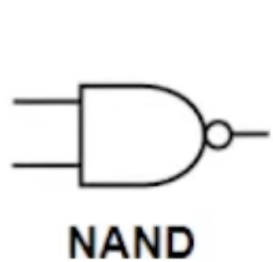
Boolean complete Logic gates of genetic circuits



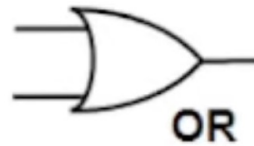
Input	Output
I	F
0	1
1	0



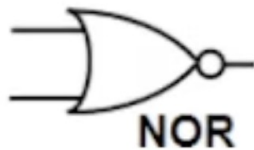
Inputs		Output
A	B	F
0	0	0
1	0	0
0	1	0
1	1	1



Inputs		Output
A	B	F
0	0	1
1	0	1
0	1	1
1	1	0

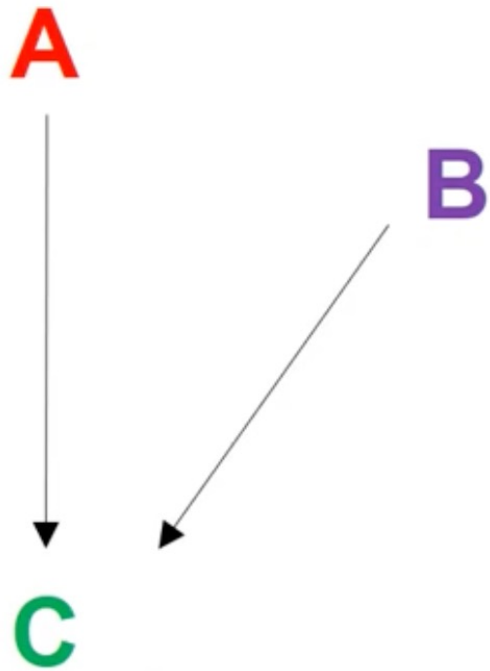


Inputs		Output
A	B	F
0	0	0
1	0	1
0	1	1
1	1	1



Inputs		Output
A	B	F
0	0	1
1	0	0
0	1	0
1	1	0

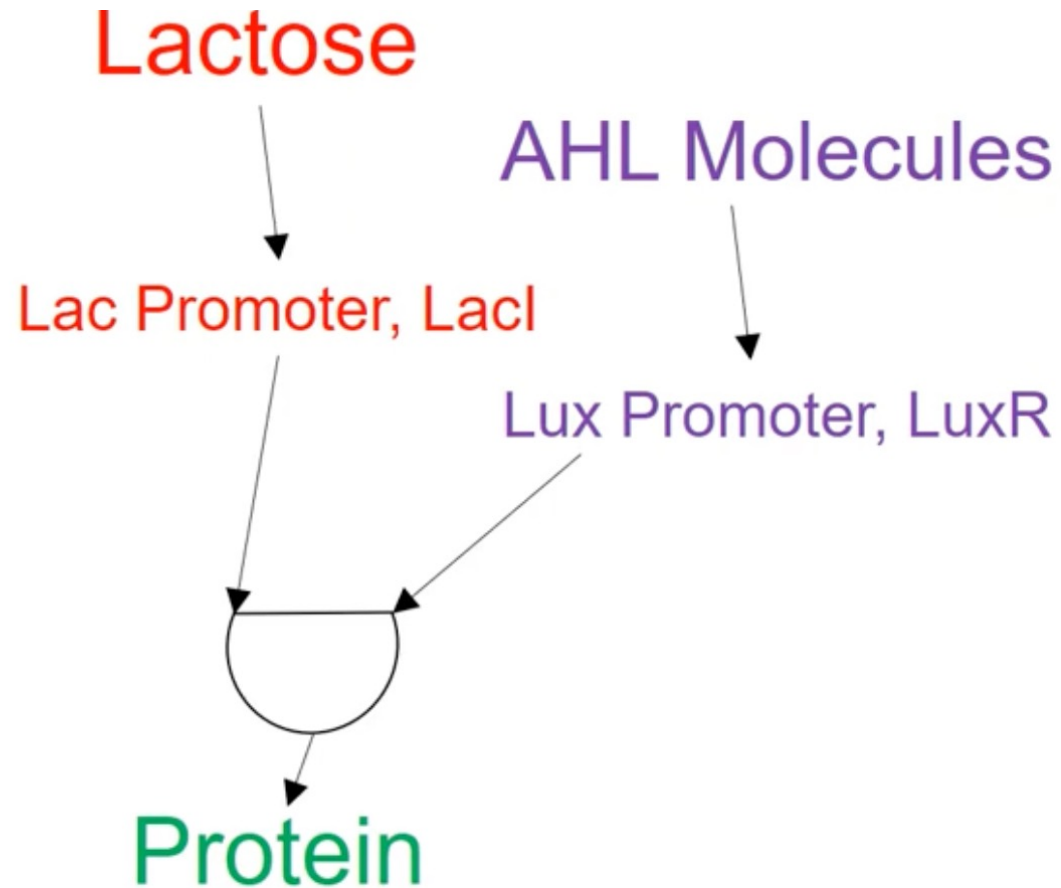
AND Gate: Two inputs



An **input (A)** and another **input (B)** is required to produce the **output (C)**

**Can be used to
integrated multiple
signals, prevent false
starts**

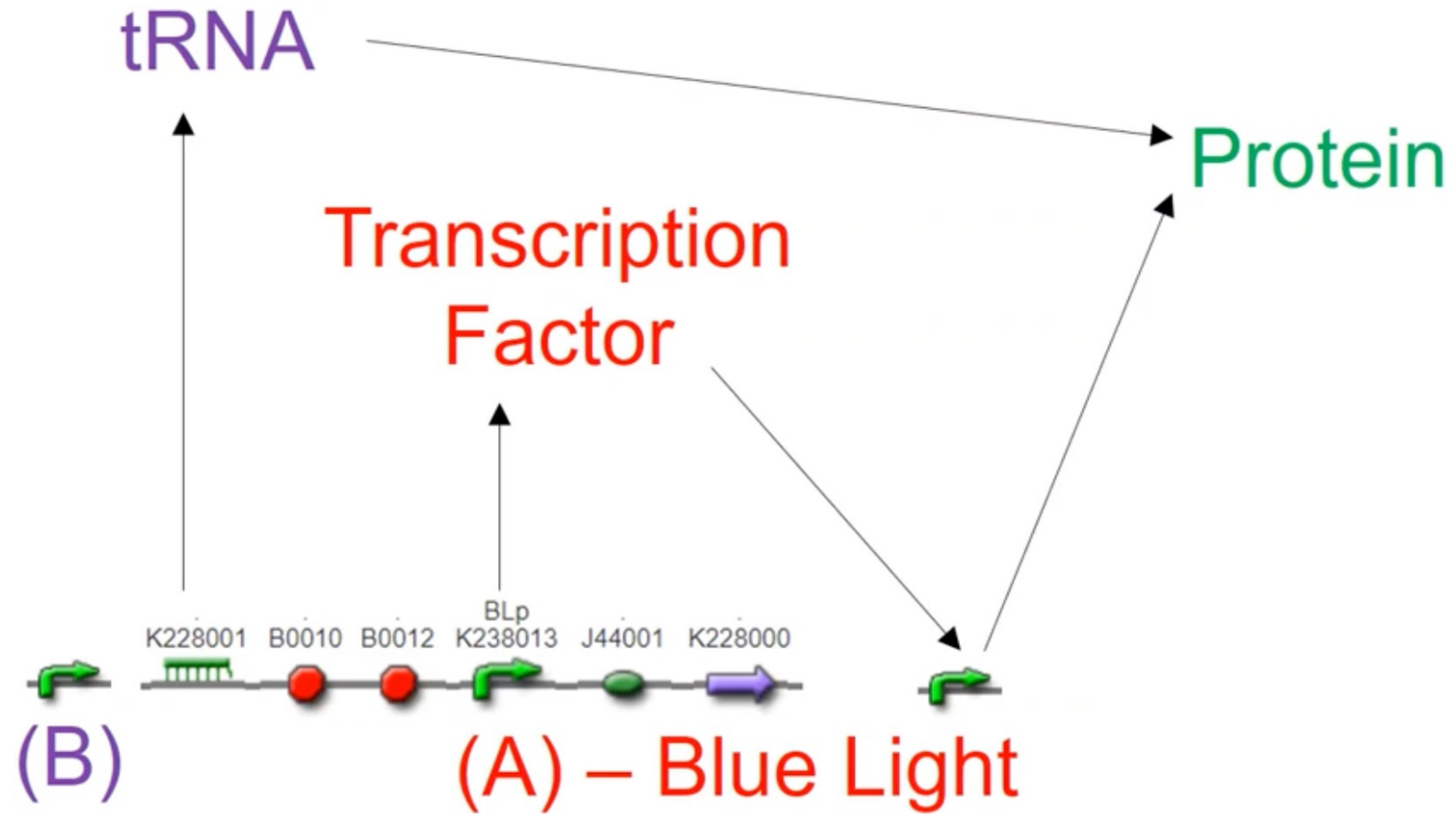
AND Gate: Two inputs



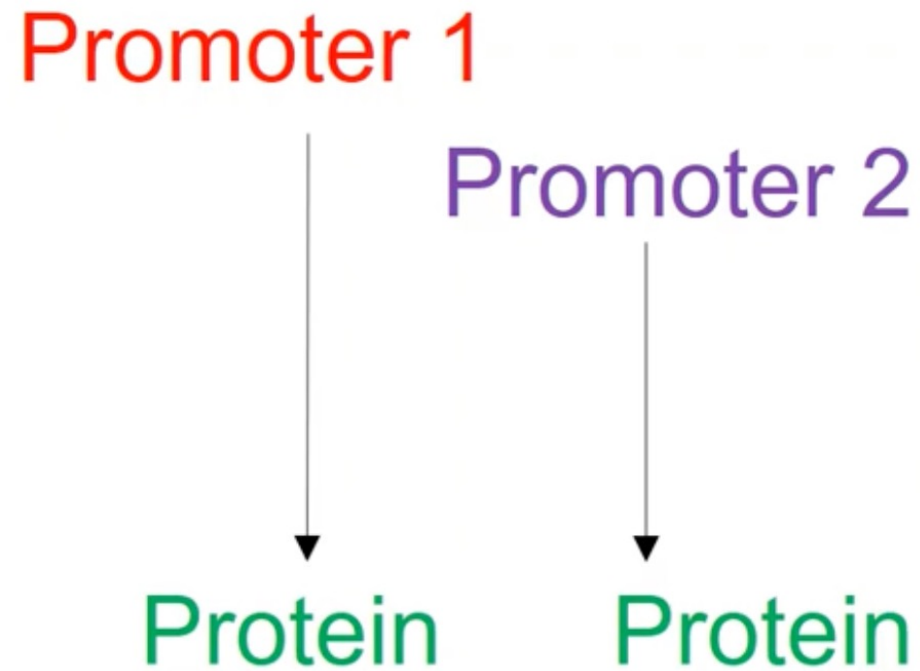
**Can be easily made
using a Hybrid Promoter:**
Contains multiple operator
/ repressor binding sites

**Only a true AND Gate if
both signals are required**

AND Gate: Two inputs



OR gate:

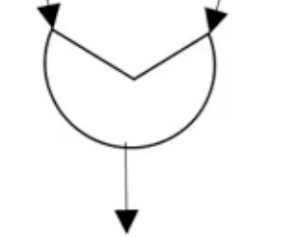


Can be easily made by
combining different
promoters and the same
genes in **parallel**

OR gate

Promoter 1

Promoter 2

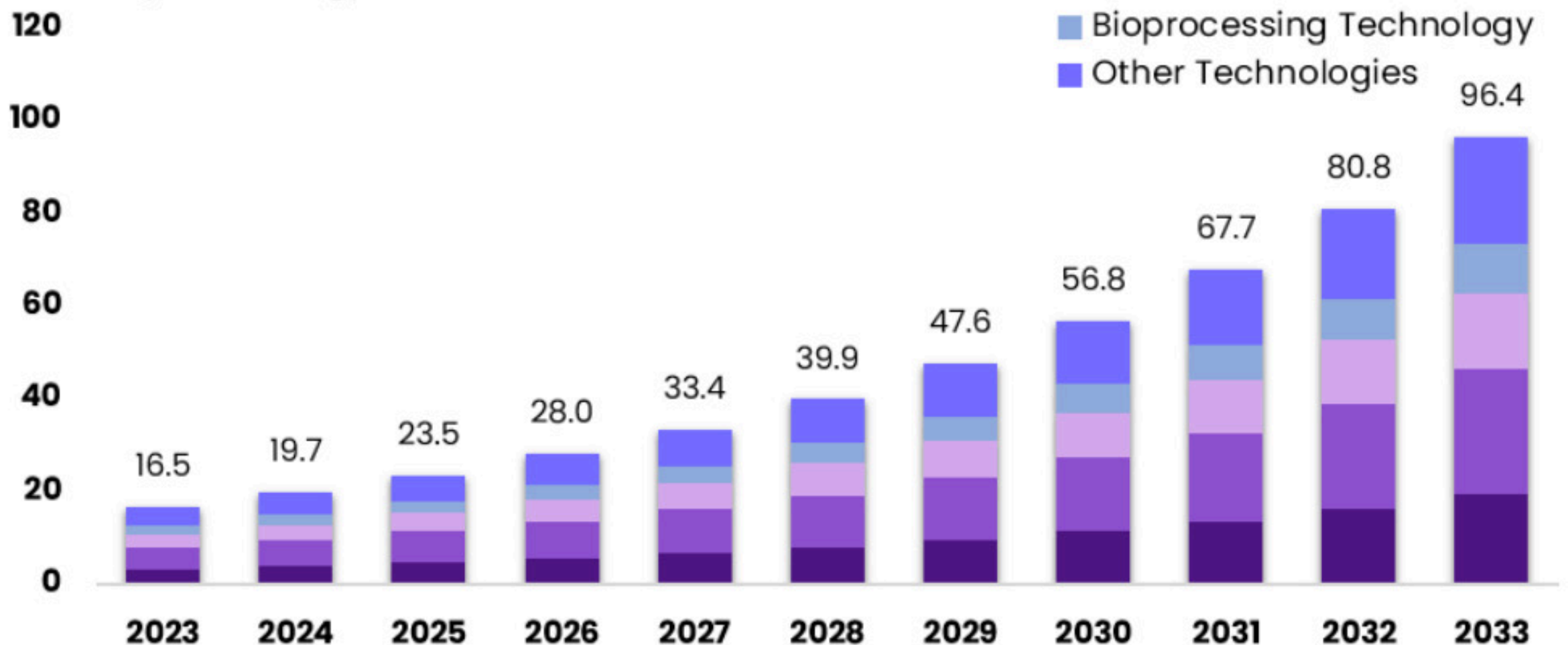


Protein

Can be easily made by
combining different
promoters and the same
genes in **parallel**

Global Synthetic Biology Market

Size, by Technology, 2023-2033 (USD Billion)



The Market will Grow
At the CAGR of:

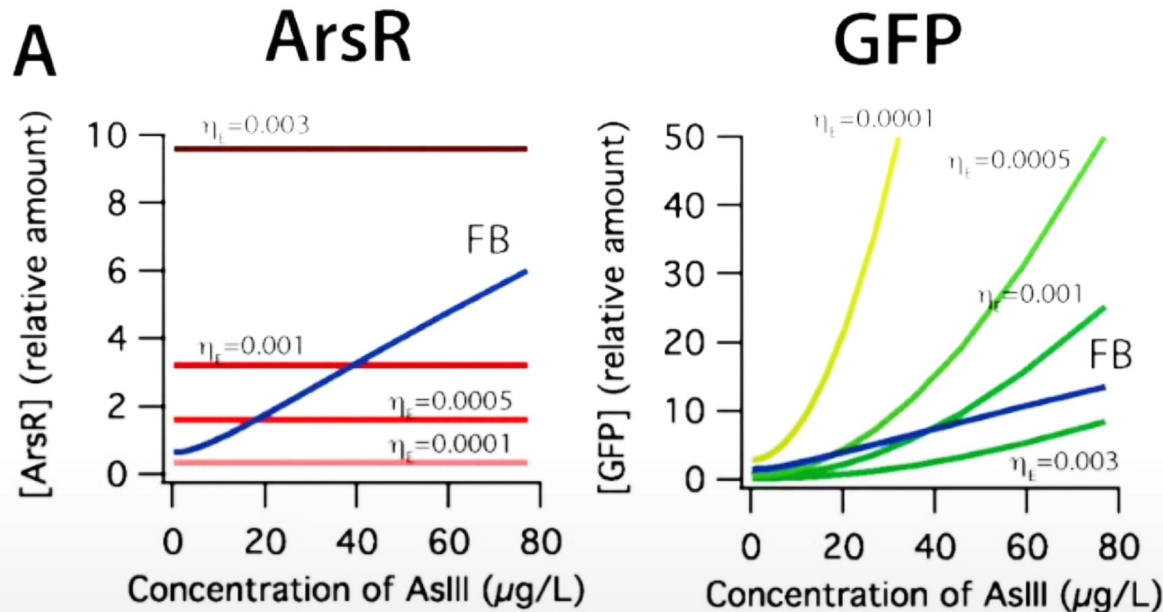
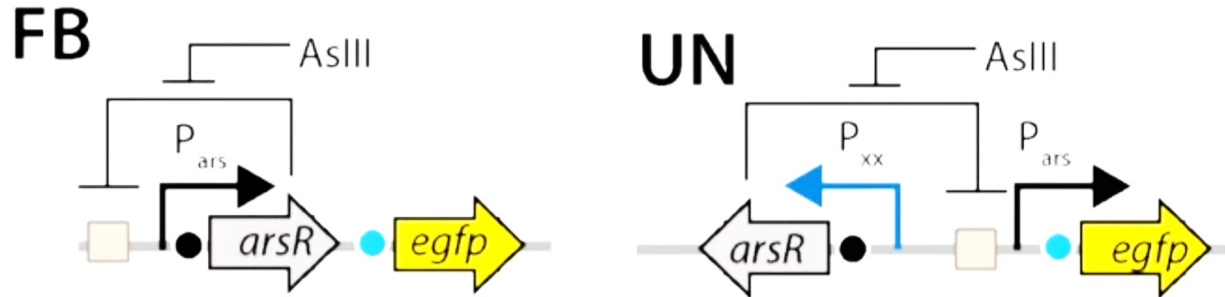
19.3 %

The Forecasted Market
Size for 2032 in USD:

\$96.4B

market.us
ONE STOP SHOP FOR THE REPORTS

Functioning of a DNA circuit



– Doxycycline



+ Doxycycline

